

Generating Uncertainty Sets Using Conformal Prediction for Robust Optimization in Minimum Cost Flow

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Abstract

Uncertainty presents a significant challenge in optimization, particularly in problems where unpredictable costs influence decision-making. Robust Optimization (RO) addresses this issue by modeling uncertainty through uncertainty sets, which ensure solutions hold under worst-case scenarios, with the model’s success depending directly on the accuracy of these sets. This research explores the application of conformal prediction to construct uncertainty sets for RO, an area not yet extensively investigated. The study specifically tests split and full conformal prediction within a robust optimization minimum cost flow problem. These methods are compared against traditional techniques, including interval-based and normal-based ellipsoidal uncertainty sets. Experiments were conducted across various network structures and under different error distributions, encompassing normal, skewed, and bimodal cases. The results indicate that conformal prediction uncertainty sets perform comparably to traditional methods, exhibiting slight variations in solution quality and coverage across different scenarios. While the conformal prediction uncertainty sets generally demonstrated tighter cost variability, their median performance did not consistently surpass that of traditional approaches. Notably, the full conformal models exhibited inconsistencies in coverage, a finding that warrants further investigation into the best method for generating an uncertainty set via full conformal prediction. The findings also highlight that network characteristics, particularly structural properties and inherent cost distributions, may influence the effectiveness of these uncertainty sets.

Keywords: Uncertainty quantification, multi-target regression, prediction regions, conformal prediction, constrained optimization

1. Introduction

Effectively managing uncertainty is a fundamental challenge in mathematical optimization. The traditional deterministic optimization framework assumes all parameters are known exactly and never change. This naive assumption can result in highly infeasible nominal solutions. As reported in [Ben-Tal and Nemirovski \(2000\)](#), small, $\approx 0.01\%$, perturbations to problem parameters in 13 of 90 NETLIB Linear Programming problems resulted in constraint violations of 50% up to 210,000%. Therefore, handling uncertainty is critical,

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especially when exact parameter values may be unknown or subject to variation. In this work, we consider a subset of such problems for which it is reasonable to assume the cost function parameters are uncertain.

Robust Optimization (RO) addresses this challenge of uncertain parameters by re-framing the optimization problem as, typically, a bi-level program (Ben-Tal et al., 2009), called the robust counterpart. The original minimization problem is augmented by an inner, bounded maximization problem representing the “adversarial” nature of uncertainty. This inner problem drives parameters to their worst-case (for the decision maker) realizations within the prescribed uncertainty set. Parameterizing this uncertainty set is one of the two main RO challenges (Gorissen et al., 2015; Ben-Tal et al., 2009), the second main challenge being deriving a computationally tractable robust counterpart. The success of this RO approach strongly depends on the size and accuracy of the utilized uncertainty sets. In this research we examine the application of Conformal Prediction (CP) to parameterize these uncertainty sets.

Conformal Prediction (CP) is a model-agnostic framework for quantifying predictive uncertainty by constructing finite-sample valid prediction sets with minimal distributional assumptions (Vovk et al., 2005; Angelopoulos and Bates, 2021). Rather than relying on knowledge of the underlying distribution (e.g., normal), CP leverages the notion of sample exchangeability to calibrate prediction regions using past data, through conformity scores that measure how unusual a new observation is relative to observed samples. This framework results in prediction sets that achieve a user-specified coverage level (e.g., $1 - \alpha$) using a finite number of samples, regardless of the underlying data distribution (Lei et al., 2018; Barber et al., 2021). A key advantage of CP is its flexibility: it is a machine learning “wrapper”, i.e., a post-processing step that can be layered on top of any machine learning predictive model, including regression or classification, to generate statistically valid prediction intervals from that model’s outputs. In this work, we utilize CP to construct data-driven uncertainty sets for usage within robust optimization, an approach that has not been widely explored prior to our research.

In the following sections, we provide a deeper foundational background on both robust optimization and conformal prediction.

1.1. Robust Optimization

Robust optimization incorporates uncertainty directly into the optimization problem by optimizing against worst-case realizations of uncertain parameters, thereby enabling risk-averse decision-making. Unlike stochastic optimization approaches, which assume a known probability distribution, robust optimization avoids distributional assumptions and protects against infeasibility and worst-case outcomes within a prescribed uncertainty set.

The **uncertainty set**, denoted as $\mathcal{U}$, mathematically characterizes the range of all admissible values for the uncertain parameters. Note the intentional usage of the word *admissible*, the uncertainty set is not the range of all possible values for the uncertain parameters but the decision-maker defined range they wish to protect against. Formally, a robust optimization formulation generates an optimal solution x^* such that,

$$x^* = \arg \min_{x \in \mathcal{X}} \sup_{\Delta \in \mathcal{U}} c(x, \Delta), \quad (1)$$

where $c(x, \Delta)$ denotes the objective under uncertain parameters Δ , subject to a feasible region $\mathcal{X}$ and an uncertainty set $\mathcal{U}$ for the uncertain parameters Δ . The uncertainty set within a robust optimization formulation are meant to act as a “knob” to adjust the risk-averse nature of a decision, with larger uncertainty sets associated with a higher degree of risk-aversion. Thus, RO minimizes the worst-case outcome with respect to the given uncertainty set $\mathcal{U}$.

This research focuses on equipping a robust optimization problem with uncertainty sets parameterized via CP. Towards that end we utilize a robust optimization counterpart as an example problem. To make this formulation concrete, we focus on the robust minimum cost flow problem, a canonical example in network optimization, as our test case.

Robust Minimum Cost Flow Problem:

$$\min_{x \in \mathbb{R}_+^{|E|}} \max_{\Delta \in \mathcal{U}} \left\{ (c + \Delta)^\top x \right\} \quad (2)$$

Subject to:

$$x_{i,j} \geq 0, \quad \forall (i, j) \in E \quad (3)$$

$$\sum_{j:(i,j) \in E} x_{ij} - \sum_{j:(j,i) \in E} x_{ji} = \begin{cases} d_i & \text{if } i = \text{source,} \\ -d_i & \text{if } i = \text{sink,} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

where

- $c^\top x$ represents the nominal cost of the flow along arcs.
- Δ is the vector of deviations within the interval uncertainty set $\mathcal{U} = [-\delta, \delta]$, capturing uncertainty in arc costs.
- E is the set of edges flow can travel along.
- x_{ij} is a continuous decision variable indicating the flow on arc (i, j) .
- d_i represents the supply (+) or demand (-) at each node i .

The formulation in (2) can be simplified in certain cases to a minimization only problem by solving the dual of the inner problem.

Historically, advances in RO have been linked to the evolution of uncertainty set construction, beginning with the conservative interval sets of [Soyster \(1973\)](#), advancing to less conservative ellipsoidal sets via [Ben-Tal and Nemirovski \(2000\)](#) and then polyhedral sets ([Bertsimas and Sim, 2004](#)). Most relevant to our work [Bertsimas and Thiele \(2006\)](#) expand upon the theoretical foundations in [Ben-Tal and Nemirovski \(2000\)](#) presenting *data-driven ellipsoidal* uncertainty sets,

$$\mathcal{U}_{ell} = \left\{ u \in \mathbb{R}^d : \|\hat{\Sigma}^-(u - \bar{u})\|_2 \leq \sqrt{\chi_{d,1-\alpha}^2} \right\}, \quad (5)$$

where $\hat{\Sigma}^-$ is the estimated covariance matrix for u and $\chi_{d,\tau}^2$ is the τ -th quantile of a χ^2 -distribution with d degrees of freedom. Intuitively, $\hat{\Sigma}^-$ controls the shape of the ellipsoidal uncertainty sets and $\chi_{d,\tau}^2$ controls the size of the ellipsoidal uncertainty set. Such constructions rely on asymptotic normality assumptions, which may be violated in practice.

1.2. Conformal Prediction

Conformal prediction (CP) was first introduced in [Gammerman et al. \(1998\)](#) as a method for quantifying uncertainty in both classification and regression tasks. The techniques were expounded upon in the seminal textbook by [Vovk et al. \(2005\)](#). The primary motivation for CP is to provide statistically guaranteed prediction intervals by measuring how “unusual” a new data point is compared to existing data. Thus, CP allows practitioners to quantify uncertainty without making strong assumptions about the underlying data distribution.

In a prediction setting, the construction of a prediction set is achieved by refitting the prediction model of interest with an augmented data set, which includes the original training data $\mathcal{D}_n$ and a new data tuple (x, z) , where z is a *guess* at the true observation. Following the refitting, each observation i in the augmented data set receives a (non)conformity score, S_i , which determines the level of (non)conformity between itself and other observations. In general, a conformity score can be any measurable function. For example, one popular conformity score is the absolute residual, $|y_i - \hat{y}_i(z)|$, where $\hat{y}_i(z)$ is the predicted value for y_i generated using the augmented data set (i.e., the prediction for y_i depends on our guess for z).

For each candidate value z we construct $\pi(z)$,

$$\pi(z) = \frac{1}{n+1} + \frac{1}{n+1} \sum_{i=1}^n \mathbb{I}\{S_i(z) \leq S_{n+1}(z)\}, \quad (6)$$

where $S_i(z)$ is the conformity score associated with the data pair (x_i, y_i) and $S_{n+1}(z)$ is the conformity score associated with (x, z) . Informally, $\pi(z)$ is the proportion of observations in the augmented data set whose conformity score is less than or equal to the conformity score associated with candidate value z . Then, the conformal prediction region $\Gamma_{\text{conf}}^{(\alpha)}(x)$ for the incoming response y is,

$$\Gamma_{\text{conf}}^{(\alpha)}(x) = \{z \in \mathbb{R} : (n+1)\pi(z) \leq \lceil (1-\alpha)(n+1) \rceil\}. \quad (7)$$

In certain cases, specifically where we are performing prediction using linear models, we can perform the prediction set construction quickly through the use of closed-form solutions for the p -value change point set, as shown in [Johnstone and Ndiaye \(2025\)](#).

1.3. Related Works, Contribution, and Manuscript Organization

The effectiveness of robust optimization (RO) depends critically on the construction of uncertainty sets - sets which define the admissible perturbations of uncertain parameters ([Ben-Tal et al., 2009](#)). Classical approaches typically construct these sets either a priori, based on structural assumptions such as bounded intervals, norm constraints, or budgeted uncertainty ([Ben-Tal and Nemirovski, 2000](#); [Bertsimas and Sim, 2004](#)), or in a data-driven manner using empirical estimates of moments which lead to ellipsoidal uncertainty sets derived from the sample mean and covariance ([Bertsimas and Thiele, 2006](#)). More recent work has explored statistically motivated and learning-based approaches for uncertainty set construction, including methods based on risk measures and machine learning models ([Bertsimas and Brown, 2009](#); [Tulabandhula and Rudin, 2014](#)). While all of these approaches offer varying trade-offs between tractability and realism, they often rely on restrictive

assumptions (e.g., independence or asymptotic normality) or they lack finite-sample validity guarantees.

In this work, we instead leverage the finite-sample, distribution-free guarantees of conformal prediction to construct data-driven uncertainty sets for robust optimization. Conformal prediction provides statistically valid prediction regions under minimal assumptions, relying only on exchangeability (Vovk et al., 2005; Lei et al., 2018; Barber et al., 2021). Other uses of conformal prediction in optimization include Patel et al. (2024b) and Patel et al. (2024a). We investigate how these prediction regions can be adapted to serve as uncertainty sets within a robust optimization framework. Using a robust minimum cost flow problem as our test case, we compare several classical approaches (including interval-based uncertainty sets and moment-based ellipsoidal sets derived from sample statistics) against multiple conformal prediction based variants such as split conformal (Johnstone and Cox, 2021) and full conformal constructions. To our knowledge, the development and application of full conformal prediction based approaches have not been previously explored in this context. These methods are collectively evaluated across a range of distributional settings, including normal, skewed, and bimodal error structures, with performance assessed in terms of solution quality, variability, and empirical coverage.

The remainder of this manuscript is organized as follows: Section 2 covers our methodology in detail, including both the methods used to construct uncertainty sets as well as the network testbed we created to test the minimum cost flow problem. Section 3 includes results for all methods against all test conditions, and Section 4 offers our concluding thoughts, a summary, and potential ways future research can progress.

2. Methodology

This section outlines the methodology used to construct and evaluate uncertainty sets within a robust optimization framework. We begin by describing the construction of the uncertainty sets we explored, including both classical approaches (interval and moment-based ellipsoidal sets) and multiple conformal prediction variants, such as split conformal and full conformal methods. In particular, we introduce several novel methods for constructing RO uncertainty sets using full conformal prediction. We then detail the implementation of these approaches within a robust minimum cost flow problem, including the generation of network instances and the simulation of uncertain arc costs under a diverse range of distributional settings (normal, skewed, and bimodal). Finally, we describe the experimental design used to assess performance, focusing on solution quality, variability, and empirical coverage across different uncertainty set constructions.

2.1. Conformal Uncertainty Sets

In this manuscript, we consider multiple types of uncertainty sets within the robust minimum cost flow framework. These range from a deterministic baseline, where arc costs are fixed at their expected values, to various classical RO approaches, to data-driven constructions based on split and full conformal prediction. In particular, we explore several novel full conformal prediction-based approaches and investigate different methods for handling the multiple ellipsoidal uncertainty sets each generates. A summary of these methods is shown below in Table 1. Full details for each method are also provided. All methods assume access

Table 1: Summary of uncertainty set construction methods.

Method	Type	Key Idea
Mean-Cost	Deterministic	Uses expected arc costs (no uncertainty)
Interval	Classical RO	Independent bounds via Bonferroni intervals
Normal Ellipsoidal	Classical RO	Mean/covariance-based ellipsoid w/ χ^2 scaling.
Split Conformal (SCP)	SCP-based	Single ellipsoid from calibrated residuals
Full CP – Largest Ellipsoid	FCP-based	Creates ellipsoids for each training point and selects 95th percentile largest ellipsoid
Full CP – Worst-Case	FCP-based	Solves RO over all ellipsoids, trims worst 5% with respect to objective value, then selects most conservative
Full CP – Subsampled	FCP-based	Approximates above worst-case using randomly selected subset of samples

to historical arc cost observations, which are used to construct both classical and conformal uncertainty sets.

Classical Approaches - Uncertainty Sets:

We begin by considering several classical baseline approaches for uncertainty set construction. The simplest of these is the mean-cost model, which assumes no uncertainty and fixes arc costs at their empirical mean values computed from the training data. This deterministic formulation serves as a reference point for evaluating the impact of incorporating uncertainty into the optimization problem.

We also consider an interval-based uncertainty set, which models each arc cost independently using upper bounds derived from the training data. Specifically, these bounds are constructed from the training data using Bonferroni-corrected confidence intervals at a joint 95% confidence level, resulting in conservative per-arc bounds which account for multiple comparisons. These upper bounds define the maximum allowable perturbation for each arc in the robust optimization problem.

Finally, we include a moment-based ellipsoidal uncertainty set. This approach naively assumes arc costs follow a multivariate distribution and characterizes the ellipsoid by the sample mean and covariance. In this formulation, the ellipsoid is centered at the sample mean, with its orientation determined by the covariance matrix and its size scaled using a chi-squared quantile corresponding to a 95% confidence level.

Split Conformal Uncertainty Sets:

We next consider uncertainty sets constructed using split conformal prediction. In this approach, the available data (300 instances) is partitioned into a training set (250 instances) and a calibration set (50 instances). A predictive model (our implementation used ridge regression, but CP is model agnostic) is first fit to the training data, and then residuals are computed on the calibration set. Conformity scores are then calculated using the Mahalanobis distance between the predicted and observed responses, which captures both scale and correlation in the residuals. The resulting conformal prediction region takes the form of a single ellipsoidal uncertainty set centered at the predicted mean of a new “301st” instance, with its orientation determined by the covariance of the training set residuals

and its size calibrated using the empirical distribution of conformity scores. The resulting single ellipsoid calibrated on held-out data serves as the uncertainty set U for the robust counterpart defined in Eq. 2.

Full Conformal Prediction Uncertainty Sets:

In contrast to split conformal methods, full conformal prediction utilizes all available data when constructing prediction regions. For each of the 300 candidate instances, the model is refit using an augmented dataset, and conformity scores are computed across all observations. When using Mahalanobis distance as the conformity score, this procedure generates a collection of candidate prediction regions (i.e. a collection of ellipsoids) whose union may not be convex nor ellipsoidal in general. As a result, additional steps are required to transform these prediction regions into usable uncertainty sets for robust optimization.

This fact presents the key challenge in using full conformal prediction for RO - it produces this collection of prediction regions, not a single uncertainty set. As a primary contribution of this work, we propose and evaluate three novel aggregation strategies to transform these regions into a single, tractable uncertainty set for the robust counterpart.

In the first approach, based on work by [Johnstone and Ndiaye \(2025\)](#), we construct an uncertainty set by selecting the 95th percentile largest ellipsoid from the collection of conformal regions generated across the training data. This approach provides a computationally efficient approximation while retaining a high level of coverage.

In a second approach, we adopt a worst-case aggregation strategy. Specifically, we generate the set of 300 candidate ellipsoids, one for each instance, and then solve the robust optimization problem using each of these sets as the uncertainty set. The worst-performing 5% of solutions, in terms of the objective function value, are discarded, and the ellipsoidal uncertainty set associated with the most conservative solution among the remaining 95% is selected. This approach balances robustness against sensitivity to extreme outliers.

In our third approach, to improve computational tractability, we also consider a sub-sampled variant of the above worst-case method, where a random subset of ellipsoids is considered instead of the full set. This sampling significantly reduces computational cost while preserving much of the robustness behavior observed in the full method.

In all cases the full conformal prediction method results in a single, aggregated, ellipsoidal uncertainty set U , which is then used in the robust counterpart defined in Eq. 2.

Across both the split and full conformal approaches, ridge regression is used as the underlying predictive model. For full conformal prediction ridge regression allows for a known structure of the individual components of a conformal prediction set using the form identified in [Johnstone and Ndiaye \(2025\)](#). We assume a linear relationship between covariates and arc costs of the form $y_i = x^\top \beta + \epsilon_i$ where β is a matrix of unknown model coefficients and $\epsilon_i \sim G(\mu, \Sigma)$ where G is a general distribution with finite mean and covariance. In our setting, this assumption corresponds to modeling arc costs as linear functions of arc specific covariates, as defined in Eq. 8, where the observed cost c_{uv} is expressed as a base cost plus a linear combination of covariates and an error term. This assumption supports the use of ridge regression as the underlying predictive model and enables structured conformal prediction sets.

We next describe the construction of the network testbed used to evaluate these robust optimization uncertainty sets.

2.2. Network Generation

The problem of interest for this application is a robust minimum cost flow problem (see (2)), where uncertainty exists in the cost associated with traversing each arc. For each type of uncertainty set covered in Table 1, multiple transportation networks, (i.e., pairs of edges and nodes), were constructed.

In all of the network instances, the basic network structure was the same, comprised of 50 nodes and 246 arcs. In each instance, (as seen in Figure 1), the source node was set as node 27, the nearest node to the origin (0,0), and the sink node set as node 25, the furthest node from the origin. This consistent setup ensured comparability across all of the models and increased the chances that the shortest path in question would be meaningfully long with the potential for variation in arc selection.

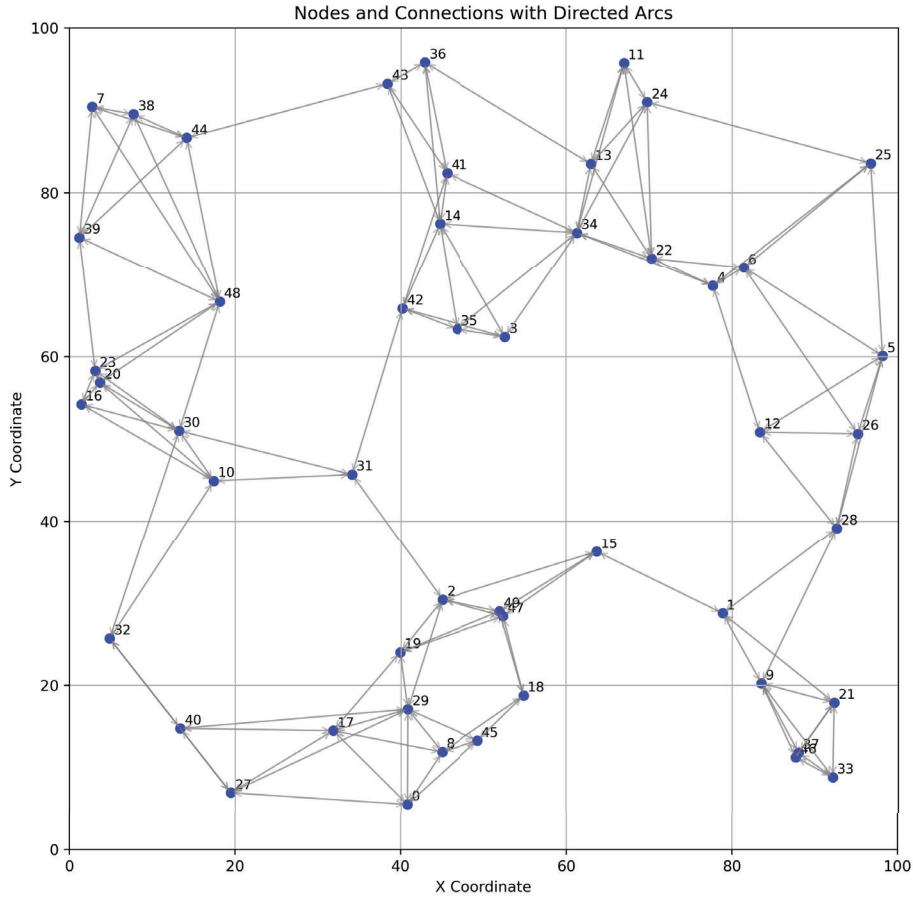


Figure 1: Network visualization showing nodes and directed arcs.

A total of 11 bidirectional networks were constructed, with the 50 nodes randomly emplaced following a uniform distribution along a bounded two-dimensional axis defined by $[x_{\min}, x_{\max}]$ and $[y_{\min}, y_{\max}]$. In this case, a square grid 100×100 . After all 50 nodes were generated, each node was connected to its three nearest neighboring nodes, ensuring the network maintained meaningful sparsity and connectivity.

In the event that an arc was generated in one direction but not the other based on the placement of the nodes, the mirroring arc was constructed in the reverse direction so that the two connected nodes could be traversed in either direction. Each direction of these bi-directional arcs were treated as unique for cost assignment purposes later. By constructing the network arcs in this way (using nearest neighbors), it forced the network to create complex source-to-sink flows paths for a particular sparsity level. All generated networks were then checked to ensure feasibility of source-to-sink flow.

2.2.1. ARC COST GENERATION

The assignment of costs to the arcs within the network was calculated by a summation of base arc cost terms, the contributions of global covariates, and a distributional error term. The base costs for the arcs were sampled uniformly on the range $[5,15]$. For each instance, three global covariates were generated, denoted by $\theta = (\alpha, \beta, \gamma)^T$, where α and β are discrete variables taking various values to represent low, mid, and high levels, respectively, and γ , is a continuous variable sampled from a uniform distribution between 0 and 1. Each level of α and β was associated with specific fixed values: $\alpha_{\text{low}} = 15$, $\alpha_{\text{mid}} = 25$, $\alpha_{\text{high}} = 35$, and similarly, $\beta_{\text{low}} = 15$, $\beta_{\text{mid}} = 25$, $\beta_{\text{high}} = 35$. The value of γ was scaled by a multiplier of 10 to reflect its influence on observed costs. Each arc in the network was assigned a unique three-element weighting vector, w_{uv} , generated via a Dirichlet distribution (such that weights sum to one), to scale the influence of the three covariates uniquely to each arc.

To correctly simulate variability and potential shifts in conditions, an error term, e_{uv} , was also included in the cost calculations. The error term values were drawn from various distributions which represented interpolations between a unimodal Gaussian (with $\mu = 0$, $\sigma = 3$, and skewness parameter $\kappa = 0$) and two target distributions: 1) a skewed Gaussian distribution (with $\mu = 0$, $\sigma = 3$, and skewness parameter $\kappa = 15$) and 2) a bimodal Gaussian distribution ($\mu_1 = -5.4$ and $\mu_2 = 5.4$, with $\sigma_1 = \sigma_2 = 3$). We interpolate between a simple Gaussian distribution and these more complex distributions using uniform steps in Earth Mover’s Distance (EMD). Examples of these distributions across multiple configurations are shown in Figure 2 and Figure 3.

Therefore, the observed cost for each arc (u, v) was calculated as the sum of the base cost (c_{uv}^{base}), the contribution of the weighted covariates ($\theta = (\alpha, \beta, \gamma)^T$), and the error term (ε_{uv}), as shown in Eq. 8.

$$c_{uv} = c_{uv}^{\text{base}} + w_{uv}^\top \theta + \varepsilon_{uv} \quad (8)$$

Though statistically improbable, the formulation also ensured non-negativity by setting costs to zero if the resulting calculated value was less than zero.

Out of the eleven network configurations, Network 1 was used to compare the performance of various RO frameworks parameterized via CP (see Table 1), to explore how various uncertainty set configurations would perform when exposed to varying distributional assumptions. This approach allowed us to explore how sensitive Robust Optimization is to various uncertainty set configurations, especially as those configurations are exposed to data that deviates from their assumptions.

In the case of Network 1, 310 instances were generated to evaluate the variability of observed costs. The first 300 instances served as the training set (for split CP the training

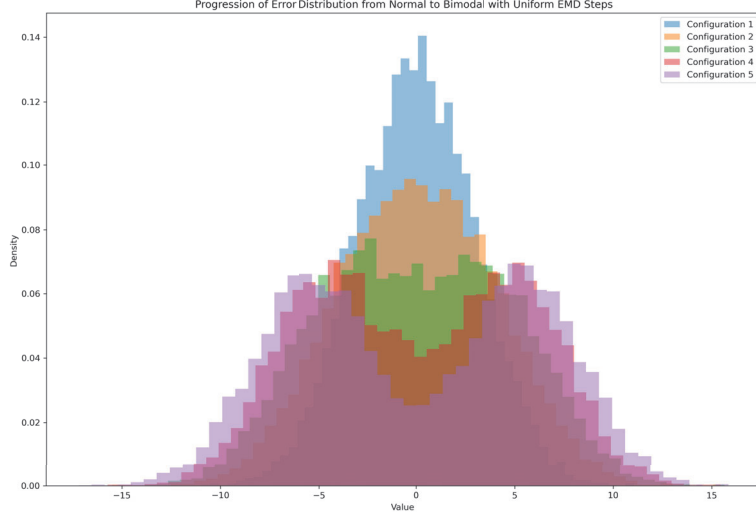


Figure 2: Progression of error distribution from a normal distribution to a bimodal distribution with uniform Earth Mover’s Distance (EMD) steps.

and calibration set) while the remaining 10 instances were reserved for testing purposes. While the network configuration was fixed, the global covariates were independently sampled, and the arc costs were recalculated for each of the instances to ensure diversity in the dataset.

For the remaining network configurations, e.g., Networks 2 through 11, 350 instances were generated for each configuration to evaluate the variability of the observed costs and the coverage guarantees of each methodological approach. Similar to Network 1, each of these networks had 300 instances used for training and calibration, with the remaining 50 used as test instances to inspect coverage performances across the various uncertainty set development methods. A detailed explanation of this network construction is shown in Figure 4.

3. Results

This research evaluated the performance of conformal prediction (CP) generated uncertainty sets in robust optimization (RO) by comparing them to traditional methods like interval-based and normal-based ellipsoidal uncertainty sets. Experiments were conducted across various network structures and error distributions (normal, skewed, and bimodal) in a robust optimization minimum cost flow problem. Our analysis focused on three key metrics: solution quality, variability, and empirical coverage, with particular attention to how the various RO uncertainty set methods (see Table 1) perform under increasingly non-normal error distributions.

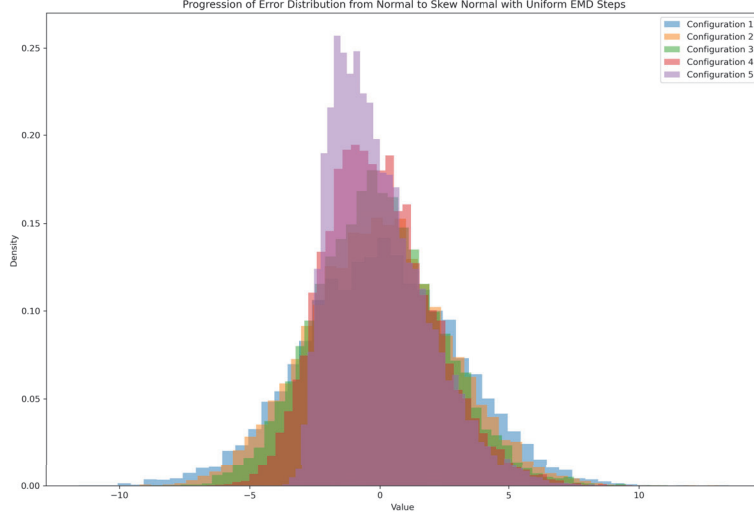


Figure 3: Progression of error distribution from a normal distribution to a skewed normal distribution with uniform Earth Mover’s Distance (EMD) steps.

3.1. Solution Quality under Skewed Error Distributions

Under skewed error distributions, the benefit of modeling covariance becomes increasingly apparent. In the near-normal case (Figure 5), both the full conformal and normal-ellipsoidal models, each of which explicitly account for covariance, produced solutions with tighter cost distributions (narrower boxes with shorter whiskers in the box and whisker plot) than the simpler interval and mean-cost models.

As the error distribution became highly skewed (Figure 6), this advantage grew more pronounced. The conformal methods (both full and split) demonstrated superior control over cost variability, evidenced by their significantly tighter whiskers. In contrast, the normal-ellipsoidal method, which assumes normality in the error term, failed to adapt and exhibited the widest cost distribution. Interestingly, the interval-based model performed better than anticipated in this high-skew regime, suggesting that a simple, covariance-blind, worst-case modeling approach can be surprisingly effective when distributions are heavily one-sided.

3.2. Solution Quality under Bimodal Error Distributions

The introduction of bimodal errors presented a different set of challenges. In the near-normal bimodal case (Figure 7), all methods produced solutions with similar cost profiles (in terms of box width, centrality, and whisker size), with the normal-ellipsoidal model slightly underperforming.

However, as the bimodality becomes more extreme (Figure 8), a clear trade-off emerges. The split conformal model proved exceptionally adept at managing this complex uncertainty,

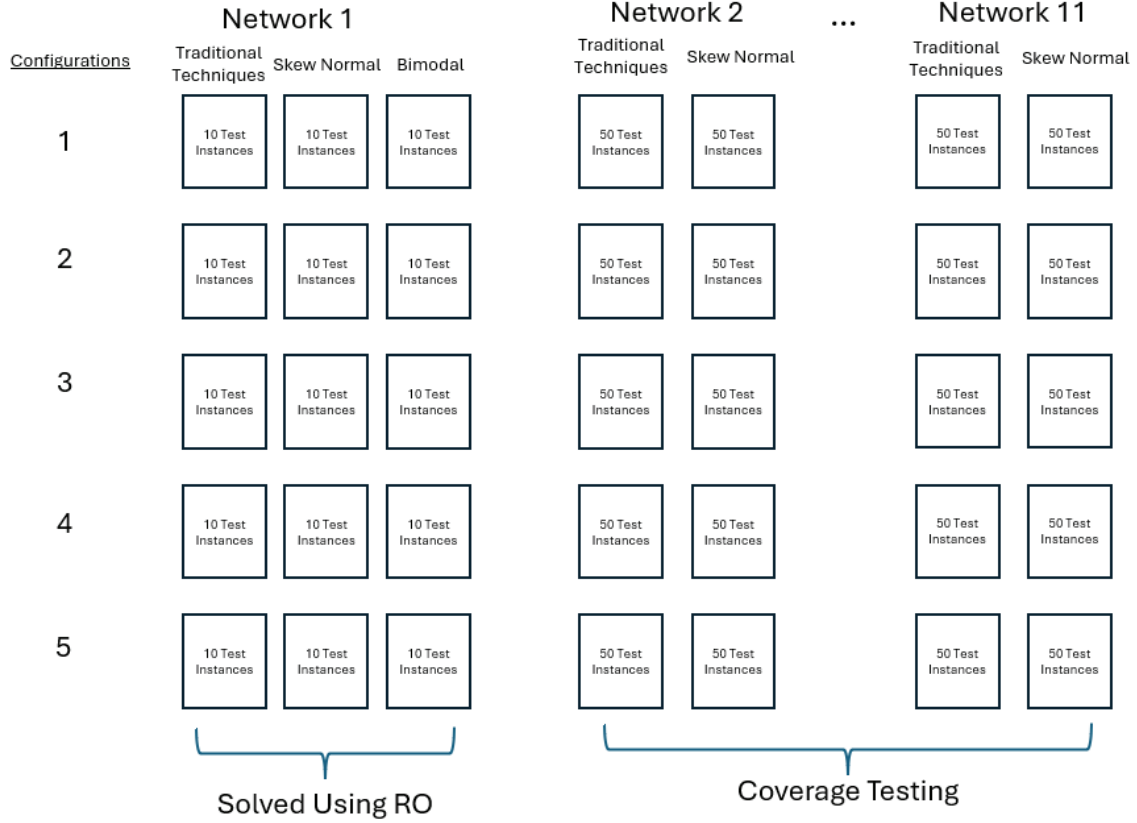


Figure 4: Network Tests and Overview

displaying the smallest inter-quartile range and, thus, the most predictable performance. Conversely, the simpler mean-cost and interval-based models, both of which ignore covariance and distributional shape, sometimes produced lower-cost solutions.

In optimization not considering uncertainty, this result would be indicative of a better performing approach. However, in our worst-case approach to (robust) optimization, a single better solution is not an indicator of sub-optimal performance. We are trying to perform better in the *worst-case*, i.e., in the tails of our performance distribution.

The small sample-size of our experiment also contributes to the result. Additionally, only looking at the distribution of performance, independent of the volume of the corresponding uncertainty sets, does not tell the full story. Future work would benefit from larger samples and additional exportation of uncertainty set volume.

Overall Cost Distribution Trends:

The analysis of cost distributions reveals two clear trends. First, the normal-ellipsoidal method consistently struggled to adapt to non-normal errors, showing deficiencies across most non-normal distributional configurations. Second, while conformal prediction models successfully reduced cost variability, they did not always produce the lowest-cost solutions. This trade-off between variability reduction and median cost is a clear example of the 'price

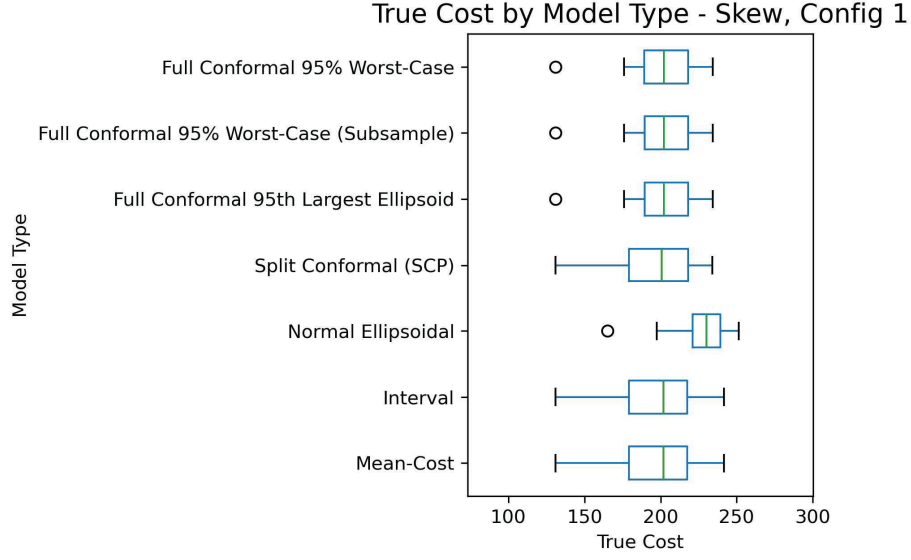


Figure 5: True Cost Distribution for near-normal skewed error distribution

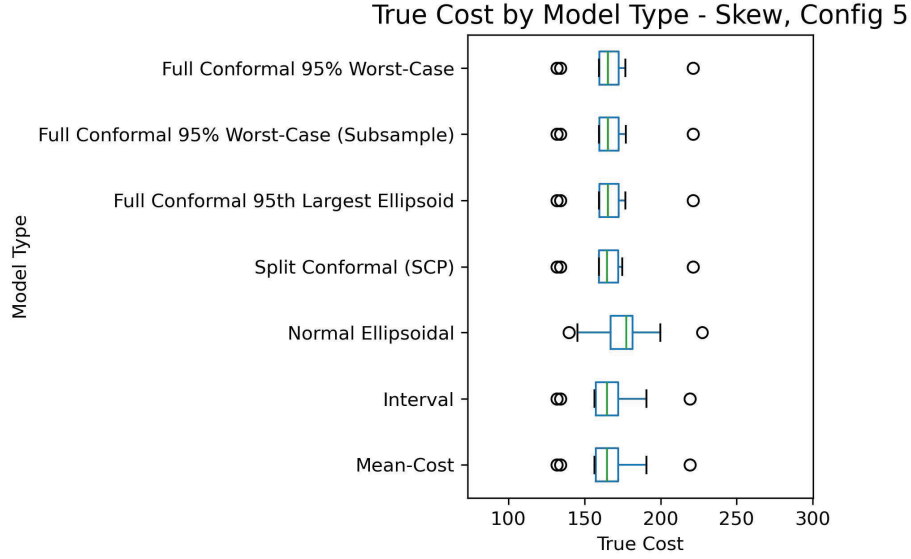


Figure 6: True Cost Distribution for highly skewed error distribution

of robustness' (Bertsimas and Sim, 2004), where protecting against worst-case outcomes can lead to more conservative, and thus not always the cheapest, solutions.

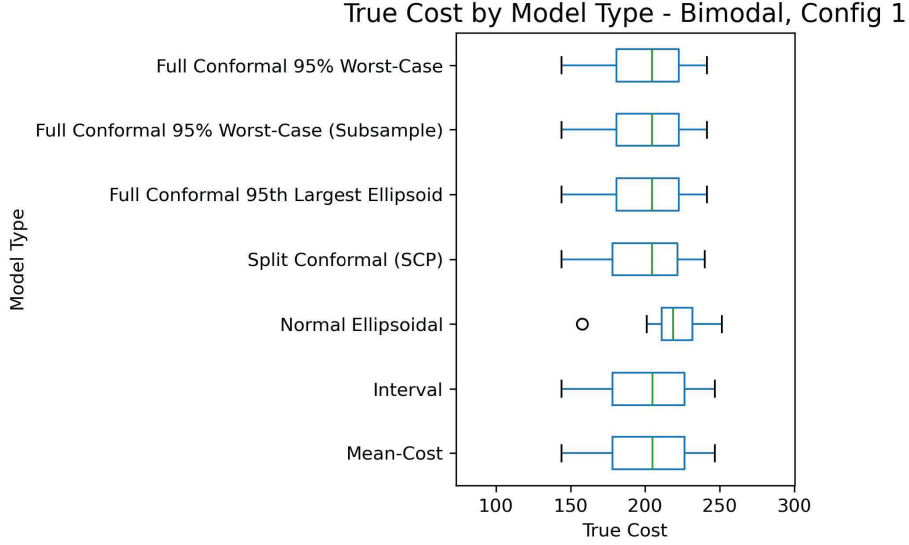


Figure 7: True Cost Distribution for nearly normal bimodal error distribution

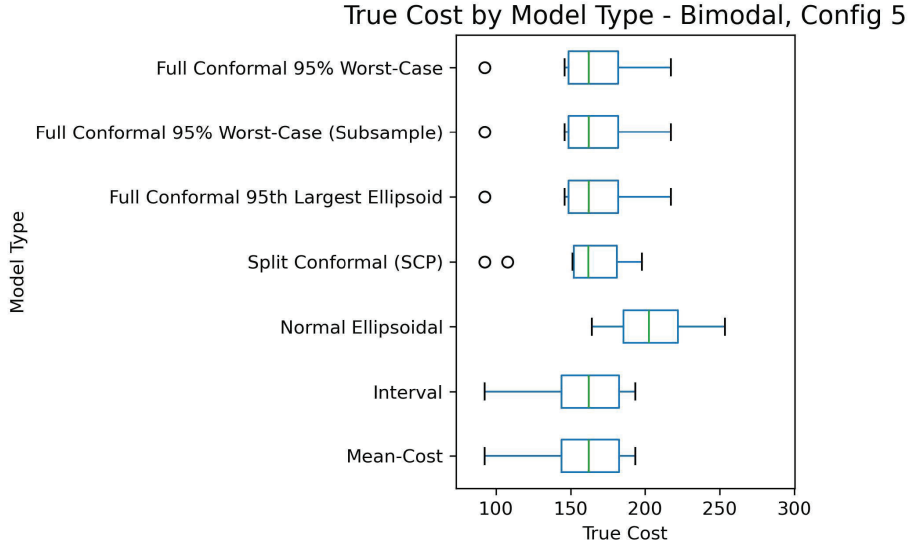


Figure 8: True Cost Distribution for highly bimodal error distribution

3.3. Path Selection Frequency Analysis

A notable feature of the cost distributions seen in Figures 5 - 8 is that while variability differed between models, median costs are strikingly similar across most models under multiple different underlying distributional assumptions. To understand this phenomenon, we analyzed the frequency with which each model selected different paths through the network. Figure 9 visualizes the paths selected for the nearly normal bimodal error distribution (a), the nearly normal skewed error distribution (b), as well as the highly bimodal error distribution

(c) and the highly skewed error distribution (d) for Network 1. The selected arc paths for the 10 test instances are all shown with line width representing the number of times the arc was selected.

The analysis revealed a small set of structurally dominant primary paths that were overwhelmingly chosen by all methods, regardless of the uncertainty model used or the underlying error distribution. Even as error distributions became highly skewed or bimodal, the optimizers consistently converged on this same small subset of routes. This similarity in selected paths indicates that for this network topology, the inherent topology of the graph has at least as strong an influence on the optimal solution as the arc cost variations. While some methods, particularly the normal-ellipsoidal, explored minor alternative routes, these deviations were insufficient to strongly shift the quality of the solution. The structural dominance of this subset of paths explains the similarity in median costs.

3.4. Coverage Analysis

While the quality and variability of the solutions is critical, a primary motivation for using advanced uncertainty models is to ensure reliability of coverage. To assess this reliability, we performed a coverage analysis to determine the proportion of test instances in which the true, observed arc costs fell entirely within the constructed uncertainty sets. The results, summarized across all 11 network configurations, reveal clear performance trade-offs against the 95% target coverage level. The results for Network 1 only are provided below in Table 2.

Table 2: Coverage Rates for Network 1

Config	Dist	Interval	Normal Ellipsoid	SCP	95th Ellipsoid	95% Retained Ellipsoid	Subsampled Ellipsoid
1	bimodal	0.6	1.0	0.8	1.0	1.0	1.0
1	skew	0.5	1.0	1.0	1.0	1.0	1.0
2	bimodal	0.9	1.0	1.0	1.0	1.0	1.0
2	skew	0.7	1.0	1.0	1.0	1.0	1.0
3	bimodal	0.4	1.0	1.0	0.1	0.4	0.4
3	skew	0.7	1.0	1.0	1.0	1.0	1.0
4	bimodal	0.8	1.0	0.9	0.0	0.0	0.0
4	skew	0.5	1.0	1.0	1.0	1.0	1.0
5	bimodal	0.8	1.0	0.8	0.0	0.0	0.0
5	skew	0.9	1.0	1.0	1.0	1.0	1.0

The Interval Model consistently under-covered across both bimodal and skewed distributions, failing to meet the expected 95% level. This under-coverage is likely due to its incorrect assumption of independent variations between arc costs.

In stark contrast, the Normal Ellipsoidal Model demonstrated consistently strong but overly conservative performance, achieving perfect (100%) coverage in all configurations. This over-coverage indicates the model captures more variability than necessary, resulting in overtly conservative performance even for non-normal scenarios.

The Split Conformal (SCP) Model achieved the best balance. It performed adequately across all network structures, with its empirical coverage fluctuating near the target 95%

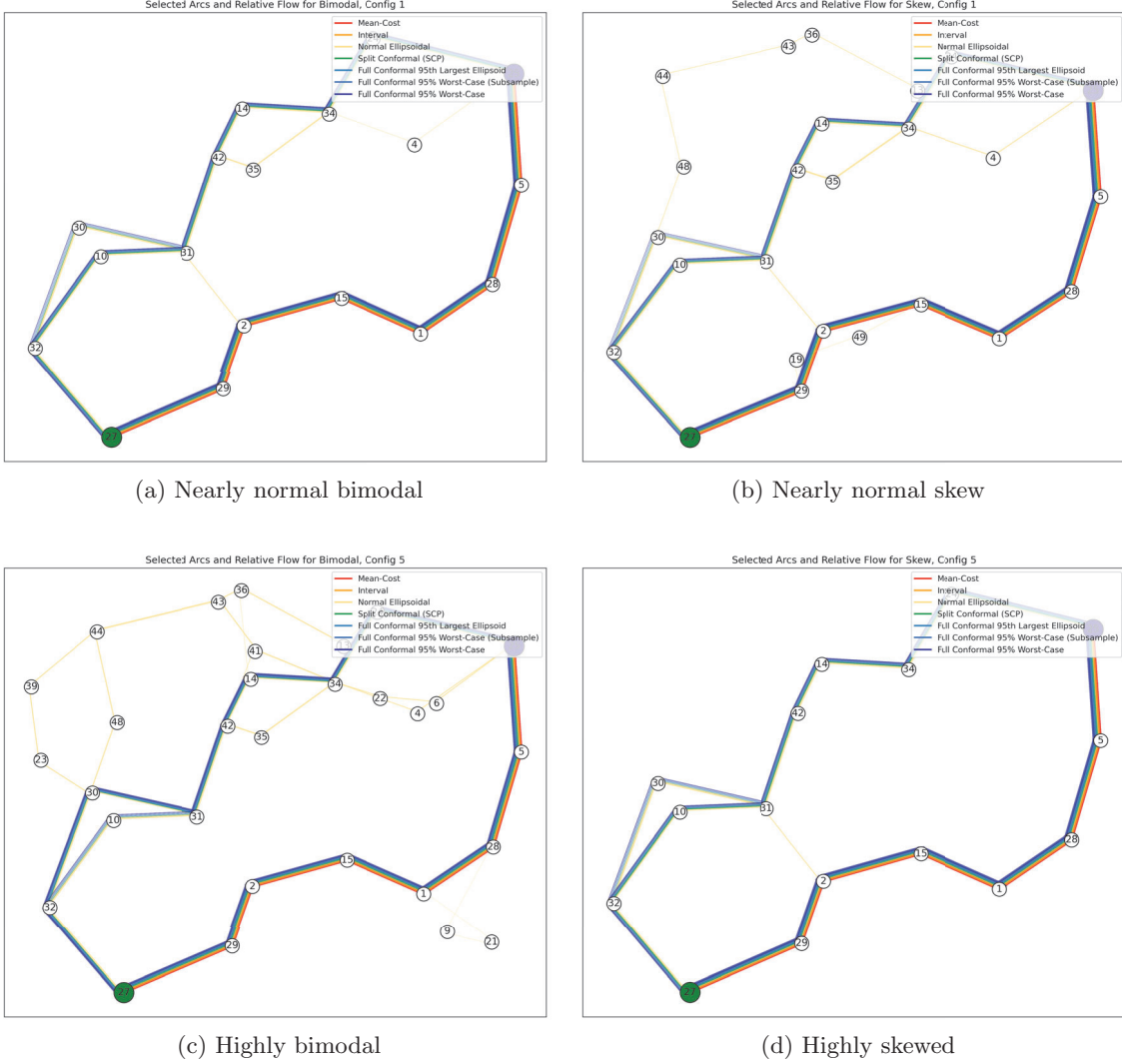


Figure 9: Visualization of selected routes by distribution. Line weight represents the number of times the arc was selected.

level. Crucially, it did not exhibit the same degradation in bimodal settings that the full conformal models did, making it a relatively stable and reliable choice.

To evaluate coverage performance more robustly we expanded our analysis beyond the initial 10-instances on Network 1. As depicted in Figure 4 we generated an additional 10 networks (Networks 2–11) each following the same error configurations as Network 1, but with 50 test instances per configuration. This increased our effective test set from 10 to 500 instances per configuration. Because the goal of this expansion was strictly coverage evaluation rather than optimization performance, the full CP heuristic models requiring objective value evaluations were excluded. Furthermore, due to computational inconsistencies

in generating full conformal ellipsoids at extreme bimodal distributions, the aggregated coverage analysis focuses exclusively on the skewed error progressions.

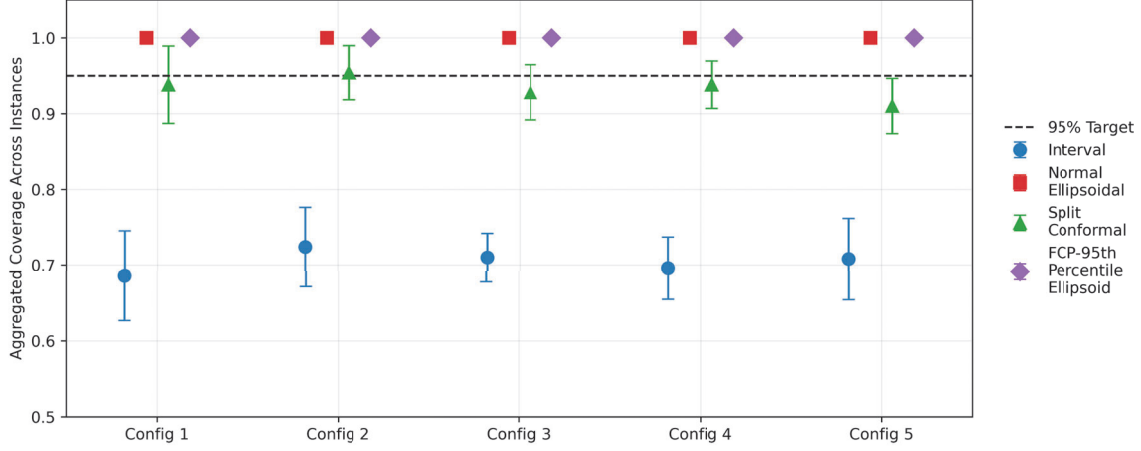


Figure 10: Empirical coverage of Networks 2-11 across increasing skewed error distributions. Error bars represent 95% confidence intervals across aggregated network instances.

As shown in Figure 10, the aggregate performance across these 500 instances mirrors the trends observed in Network 1. Specifically, the interval models consistently underperformed the more advanced techniques, achieving approximately 70% empirical coverage. The split conformal models achieved roughly 93% coverage; however, in all but one configuration, the target 95% coverage fell within the $\alpha = 0.05$ confidence interval. Both the normal ellipsoidal model and the full conformal prediction (95th-percentile) heuristic successfully achieved the target coverage level across all evaluated instances. Consistent with earlier observations, increasing the skewness of the error distribution did not noticeably impact coverage performance for any model.

The observed performance of the three full conformal models highlights the challenge of applying full CP to generate RO uncertainty sets. While full CP provides strong theoretical coverage guarantees, these guarantees apply to the entire, often jointly non-convex, prediction region it generates. Because RO requires a single, typically convex, uncertainty set, our methodology necessitated an aggregation step to select a single tractable ellipsoid. The degradation in coverage seen in bimodal settings suggests that these heuristic aggregation methods did not fully preserve the properties of the original conformal set. This points to a critical area for future work: developing aggregation strategies that better translate the theoretical guarantees of full conformal regions into practical uncertainty sets for robust optimization.

4. Conclusion

This research examined the application of conformal prediction (CP) generated uncertainty sets within the framework of robust optimization (RO), addressing a gap in the literature

regarding its direct application and providing a comparative performance assessment against traditional methods. By integrating split and full conformal prediction into a robust optimization minimum cost flow problem, the study systematically compared these novel approaches with established interval-based and normal ellipsoidal based uncertainty sets across diverse network structures and error distributions, including normal, skewed, and bimodal scenarios.

The results revealed that conformal prediction uncertainty sets performed comparably to traditional methods, showing slight variations in solution quality and coverage. Conformal prediction models generally demonstrated tighter control over cost variability. This reduction in cost variability indicates a more precise bounding of potential worst-case costs, especially under highly skewed error distributions (e.g., Figure 6) where full and split conformal methods showed significantly reduced variability compared to traditional techniques. Under conditions in which the underlying errors followed bimodal distributions, particularly in the highly Bimodal case (e.g., Figure 8), the split conformal model notably displayed the smallest inter-quartile range, highlighting its impressive ability to manage variability in extremely non-normal error terms.

However, this improved variability control did not consistently translate into superior overall optimization performance, as conformal prediction models did not always find lower-cost solutions compared to traditional approaches. For instance, in some bimodal configurations, interval and mean-cost models produced realized solutions with lower values in their inter-quartile range. This result is not surprising and aligns with the known “price of robustness,” where the enhanced protection against worst-case scenarios comes at the expense of achieving the absolute lowest objective function values.

Coverage analysis presented nuanced findings. While conformal methods are theoretically designed to provide strong coverage guarantees, observed coverage inconsistencies in full conformal models were noted, especially as bimodal error structures became more pronounced. This degradation implies that further investigation is needed to develop better heuristic aggregation approaches for full conformal methods to leverage these strong coverage guarantees. Conversely, the normal ellipsoidal model demonstrated consistently strong coverage (often 1.0) across all configurations, even non-normal ones. However, this high coverage was often accompanied by poorer solution quality, exhibiting both higher median and 75th percentile costs in nearly all configurations. The interval model, while computationally efficient, consistently underperformed in coverage, reinforcing its limitations in capturing complex cost dependencies.

The findings indicate that full conformal prediction holds potential to reduce variability in prediction sets, offer moderately good robust optimization results, and demonstrate strong validity in certain simulated distributional scenarios. However, the observed trade-offs between computational efficiency, coverage reliability, and optimization performance highlight that the advantages of full conformal models are not always substantial enough to justify their added complexity, especially when compared to the more balanced split conformal prediction, which often achieved comparable results with less computational effort. Future research should explore alternative uncertainty set constructions (e.g., minimum volume ellipsoids), investigate the impact of various network structures and covariate influences, and delve deeper into conformal prediction’s efficacy under diverse and extreme distributional shifts to truly define its role and advantages in robust optimization.

This research significantly contributes to the academic understanding of integrating machine learning-driven uncertainty quantification into network optimization under uncertainty, providing one of the first direct comparisons of conformal prediction with traditional robust optimization techniques, particularly in non-normal distributional settings. For robust logistics applications, this work represents a foundational step towards integrating data-driven machine learning into planning, even while it underscores the need for extensive further research with real-world data and model refinement before deployment.

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